

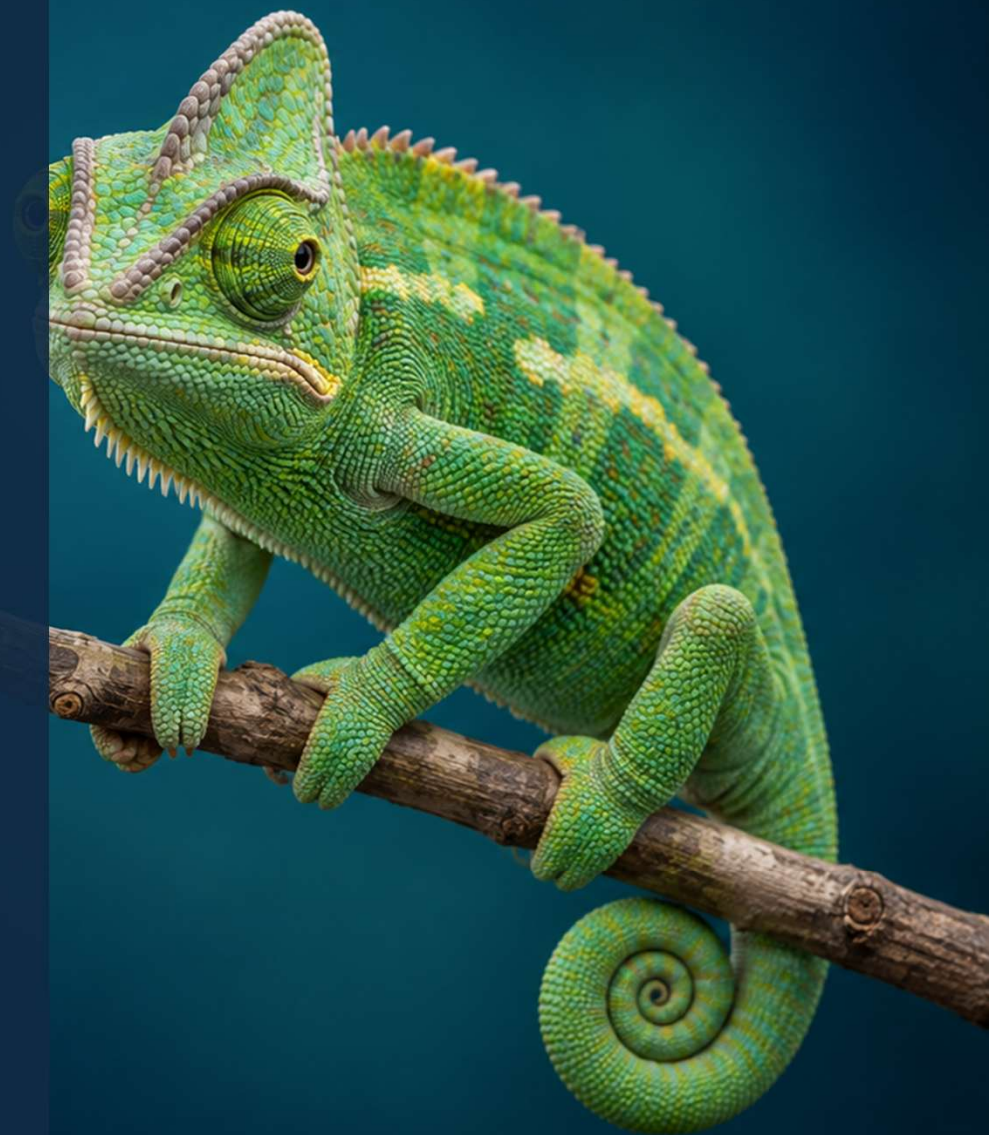
A03 • NEURAL NETWORK ZOO

CNN Chameleon

From biological neurons to visual learning

Group 4 | ITAI 2376

Neural Network Zoo



Why the CNN Chameleon?

Our guiding question

How can a computer start with a grid of numbers and eventually decide what is in an image?



We will follow the information, not just memorize the terminology.

Why a chameleon?

- Its eyes can survey different directions, giving us a natural visual analogy for examining a scene.
- The analogy stops there: a CNN does not literally “look around.” It applies mathematical filters to image data.
- That distinction helps us separate biological inspiration from the actual computation.

Why CNNs were created

The story starts with a very practical vision problem:

How can a computer recognize the same thing when it moves around in an image?

1 THE PROBLEM

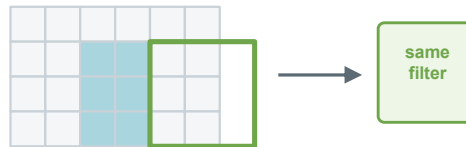
A “7” is still a 7 — even when it moves.



A basic system could treat each location as a different problem.

2 THE IDEA

Reuse the same small pattern detector.



1980 • Kunihiko Fukushima’s Neocognitron showed how layered visual detectors could recognize patterns even when their position changed.

3 THE REAL-WORLD TEST

Could it read messy human handwriting?



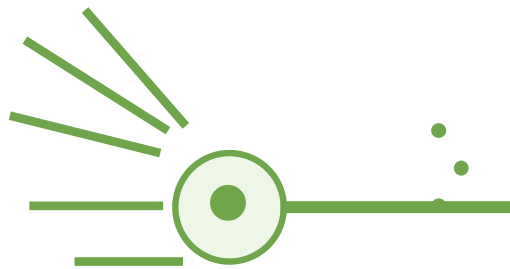
Late 1980s • Yann LeCun and colleagues used trainable convolution-style networks for handwritten ZIP-code recognition.

So CNNs grew from a simple idea: recognize a useful pattern once, then look for it anywhere.

But what is the basic unit doing the learning?
→ the artificial neuron

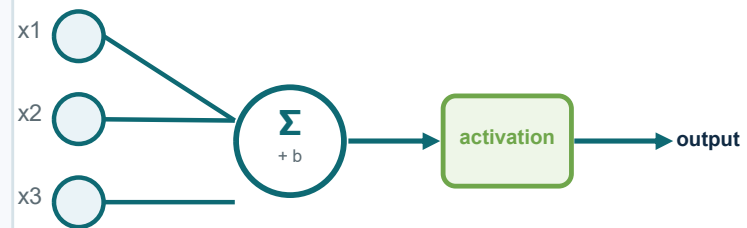
From a biological neuron to an artificial neuron

BIOLOGICAL NEURON



- Receives** Dendrites and synapses receive signals from other cells.
- Combines** The cell body integrates many incoming signals.
- Threshold** Membrane behavior determines whether an action potential occurs.
- Passes on** The axon carries the signal toward other cells.

ARTIFICIAL NEURON



- Receives** Numerical inputs enter the artificial neuron.
- Combines** Weights change importance; the neuron adds the weighted inputs plus a bias.
- Activation** An activation function transforms the combined value.
- Passes on** The output becomes input to neurons in the next layer.

Key idea Artificial neurons borrow the idea of receiving, combining, and passing signals — not the full biology of a living neuron.

How do neurons become a network? Each neuron passes its output forward. Groups of neurons working at the same stage form a layer.



Landmark: McCulloch & Pitts (1943) formalized an early mathematical neuron model.

Why a CNN adds convolution

FROM LAYERS TO CONVOLUTION

A basic neural network gives us layers.
But an image has a special structure:

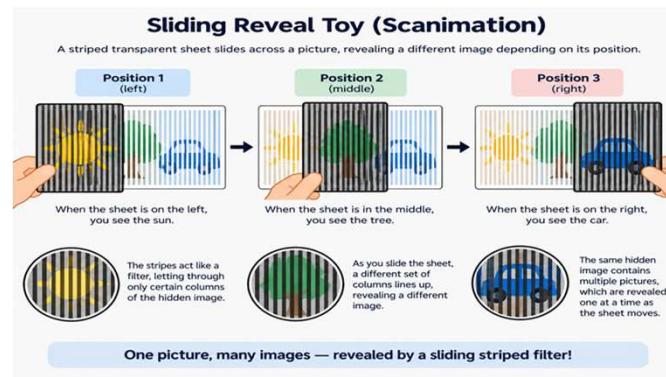
- 1 Pixels have neighbors**
Edges and textures are local patterns, not isolated pixels.
- 2 Patterns move around**
The same edge can appear anywhere in the image.
- 3 Reuse one detector**
CNNs use the same small filter across many local regions.

Deep learning = many layers
CNN = layers designed to exploit local image patterns

WHY IS IT CALLED 'CONVOLUTIONAL'? The name comes from the operation itself.

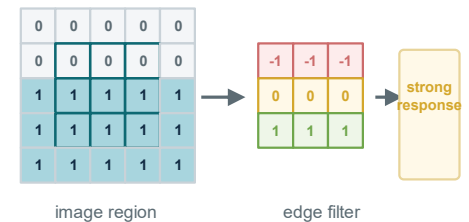
Convolution comes from the Latin root *convolvere*, “to roll together.” In image processing, a small filter is combined with one local region, then reused/rolled across the image. That repeated local operation is why the network is called convolutional.

Toy analogy: a sliding reveal toy (scanimation)



The toy gives the intuition of a moving local window. A CNN filter also moves across the image, but instead of revealing a picture, it calculates a pattern response.

Edge-detection example



When the filter lines up with an edge, the response is strong. Then the same filter is reused in the next region.

One small detector
reused many times

1980
Fukushima

CNN Chameleon: How a CNN Sees an Image

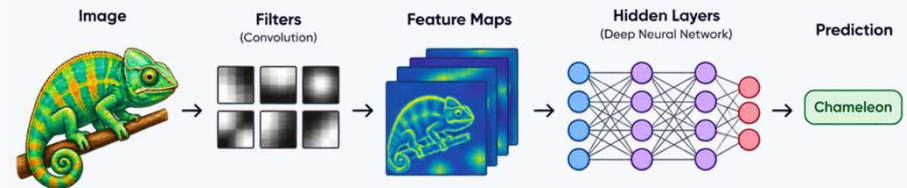
Main Components of CNN

- 1. Input Image**
The image given to a convolutional neural network.
- 2. Filters/Kernels**
Small matrices that scan across the image to detect visual patterns.
- 3. Feature Maps**
Visual maps showing exactly where specific patterns were detected.
- 4. Hidden Layers**
Process features hierarchically to gradually learn complex patterns.
- 5. Output**
Uses final learned features to generate a prediction or classification.

Understanding Convolution

A convolutional neural network is a type of neural network designed especially for working with images and visual information.

Instead of looking at the entire picture at once, a CNN examines small regions of the image and learns the important patterns locally.



From Simple to Complex Features

Hierarchical Feature Learning

Early Layers: Detect simple features like edges, lines, and colors.

Middle Layers: Combine simple shapes to recognize curves and textures.

Deeper Layers: Assemble parts into complex objects and patterns.

How the Filter Works

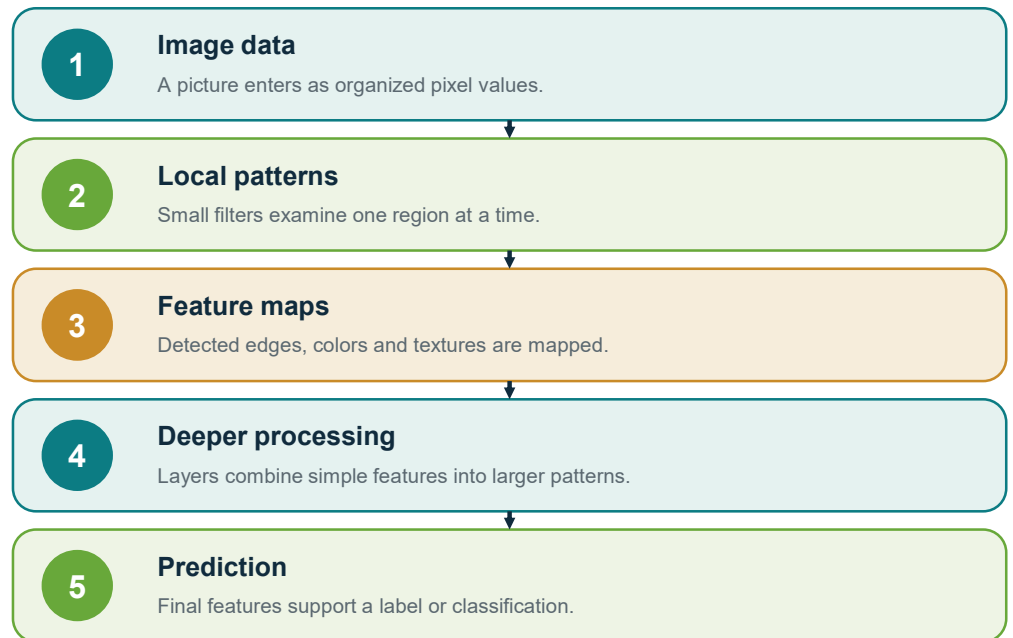
The Chameleon Analogy

Imagine a small filter scanning across an image. At each spot, it looks for specific patterns to build a feature map. This local scanning allows the CNN to learn key features efficiently without needing to analyze the entire image at once.



How the CNN Chameleon “Sees”

The eyes introduce the scanning idea; the network performs the processing.



THE KEY DISTINCTION

A chameleon gathers visual information with its eyes. A CNN scans numerical image data with mathematical filters.

Where the Analogy Works—and Where It Stops

The chameleon is a memory aid, not a literal model of a convolutional network.

WHERE IT WORKS

- 1 Different regions**
Both direct attention to separate areas of a visual scene.
- 2 Local information**
Nearby colors, edges and shapes provide useful clues.
- 3 Combined evidence**
Information from many regions contributes to recognition.

WHERE IT STOPS

- 1 Biology vs. computation**
A chameleon uses living eyes and a brain; a CNN uses calculations.
- 2 Seeing vs. numbers**
A CNN receives pixel values rather than experiencing a picture.
- 3 Scanning vs. understanding**
Pattern detection does not give the network human-like meaning.

CHAMELEON EYES

Gather information

DO NOT EQUATE THEM

The eyes are not the hidden layers

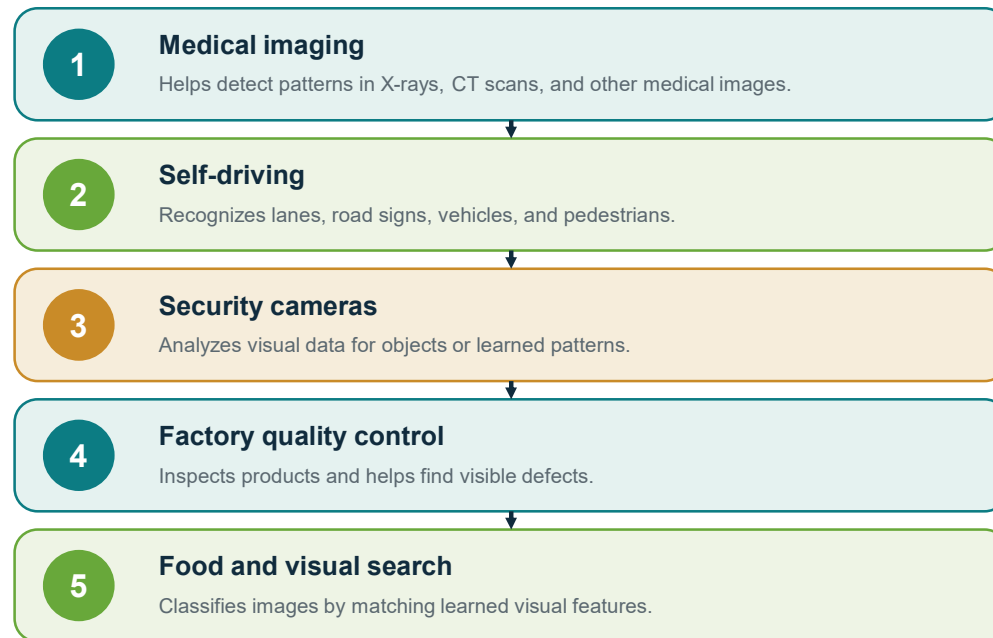
CNN HIDDEN LAYERS

Process features

Main takeaway: the chameleon explains visual scanning; filters and layers explain the actual CNN process.

Where CNNs Are Used in the Real World

CNNs are useful when a task depends on recognizing visual patterns in images or video.



WHY CNNs FIT IMAGES

CNN filters scan small regions, learn local patterns, and reuse those pattern detectors across the image.

Limitations and Final Reflection

CNNs are powerful visual pattern learners, but their results still depend on data, training, and context.

WHY CNNs ARE USEFUL

- 1 Local patterns**
They focus on nearby pixels, edges, colors, and textures.
- 2 Feature building**
Layers combine small features into larger object patterns.
- 3 Reusable filters**
The same filter can search for a pattern in many regions.

LIMITATIONS

- 1 Training data**
Biased or limited data can lead to weak predictions.
- 2 Image changes**
Lighting, blur, angle, noise, or rotation can affect results.
- 3 No human understanding**
A CNN detects learned patterns, not human-like meaning.

APPLICATIONS

Real-world use

FINAL REFLECTION

Useful, but not perfect

CNN LIMITS

Evaluate outputs

Main takeaway: CNNs turn image data into learned features and use those features for prediction, but the final output should still be checked carefully.